

The Turing Machine:-

We may visualize a Turing machine as in Fig. shown below. The machine consists of a finite control, which can be in any of a finite set of states. There is a tape divided into squares or cells; each cell can hold any one of a finite number of symbols.

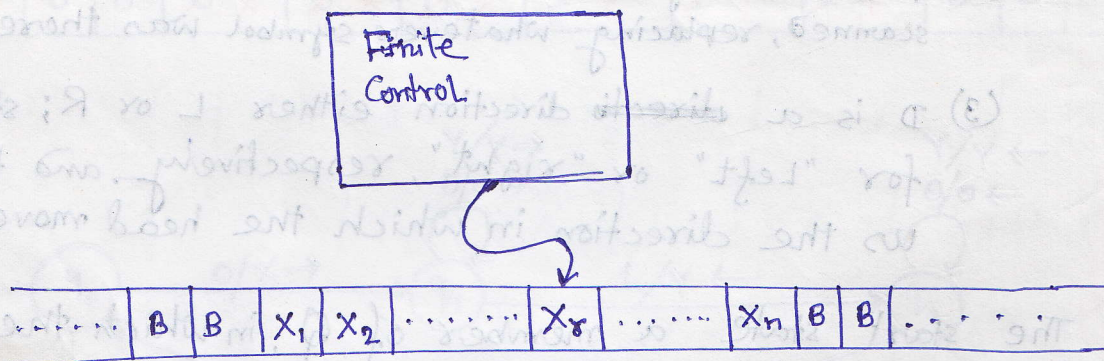


Fig: A Turing machine.

Initially the input, which is a finite-length string of symbols chosen from the input alphabet is placed on the tape. All other tape cells - extending initially to the left and right, initially hold a special symbol called the blank. The blank is a tape symbol but not an input symbol, and there may be other tape symbols besides the input symbols and the blank, as well.

There is a tape head that is always positioned at one of the tape cells. The Turing machine is said to be scanning that cell. Initially, the tape head is at the leftmost cell that holds the input.

Turing machine (TM) is similar to that used for finite Automata or PDA's. We describe a TM by the 7-tuple, -

$$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$$

whose components have the following meanings:-

Q : The finite set of states of the finite control.

Σ : The finite set of input symbols,

Γ : The complete set of tape symbols; Σ is always a subset of Γ .

δ : The transition function. The arguments of $\delta(q, x)$ are a state q and a tape symbol x .

The

The value of $\delta(q, x)$, if it is defined is triple (p, Y, D) , where:

- (1) p is the next state, in Q
- (2) Y is the symbol, in Γ , written in the cell being scanned, replacing whatever symbol was there.
- (3) D is a ~~direct~~ direction either L or R; standing for "Left" or "right", respectively, and telling us the direction in which the head moves.

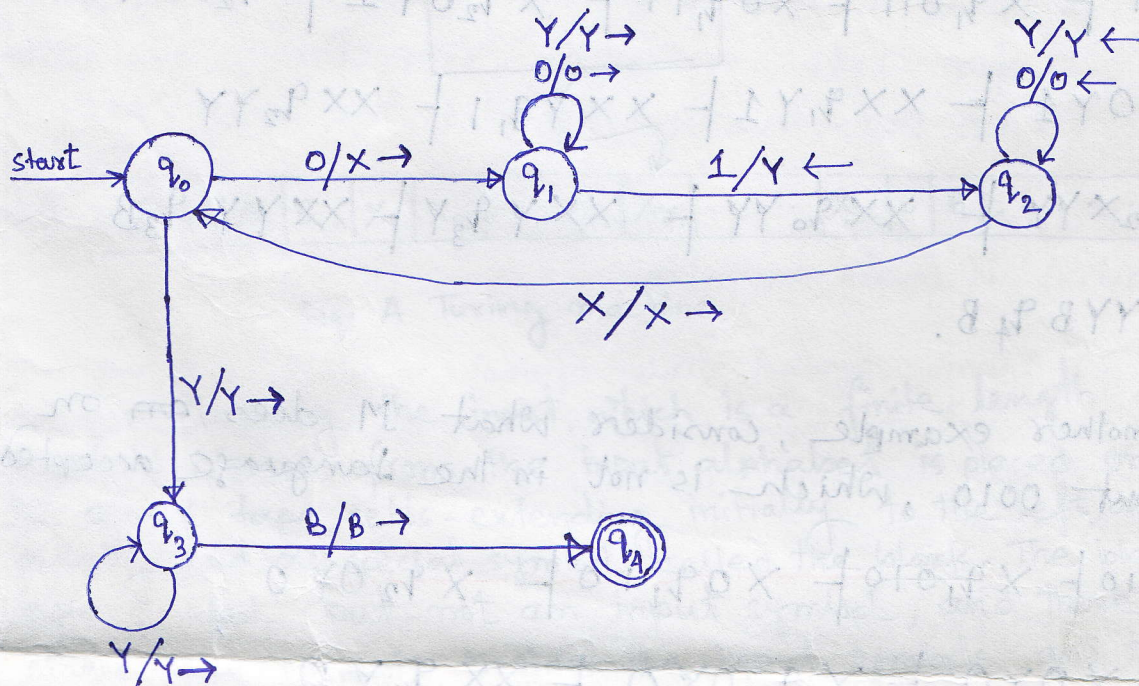
q_0 : The start state a members of Q , in which the finite control is found initially.

B : The Blank symbol. This symbol is in Γ but not in Σ ; i.e. it is not an input symbol. The blank appears initially in all but the finite numbers of ~~initial~~ initial cells that hold input symbols.

F : The set of final or accepting states, a subset of Q .

Problem 1 :-

Design a Turing machine for the language $L = \{0^n 1^n \mid n \geq 1\}$.



$M = (\{q_0, q_1, q_2, q_3, q_4\}, \{0, 1\}, \{0, 1, x, y, B\}, \delta, q_0, B, \{q_4\})$

Transition diagram for a TM that accepts L .

Transition table for the TM to accept $\{0^n 1^n \mid n \geq 1\}$

state	0	1	x	Y	B
q_0	(q_1, x, R)	—	—	(q_3, Y, R)	—
q_1	$(q_1, 0, R)$	(q_2, Y, L)	—	(q_1, Y, R)	—
q_2	$(q_2, 0, L)$	—	(q_0, x, R)	(q_2, Y, L)	—
q_3	—	—	—	(q_3, Y, R)	(q_4, B, R)
q_4	—	—	—	—	—

Instantaneous Description for Turing Machines :-

Here is an example of an accepting computation by M . It's input is 0011. Initially, M is in state q_0 , scanning the first 0 i.e M 's initial ID is $q_0 0011$. The entire sequence of moves of M is :-

$q_0 0011 \vdash x q_1 011 \vdash x 0 q_1 11 \vdash x q_2 0Y1 \vdash q_2 x 0Y1$
 $\vdash x q_0 0Y1 \vdash x x q_1 Y1 \vdash x x Y q_1 1 \vdash x x q_2 YY$
 $\vdash x q_2 x YY \vdash x x q_0 YY \vdash x x Y q_3 Y \vdash x x YY q_3 B$
 $\vdash x x YY B q_4 B.$

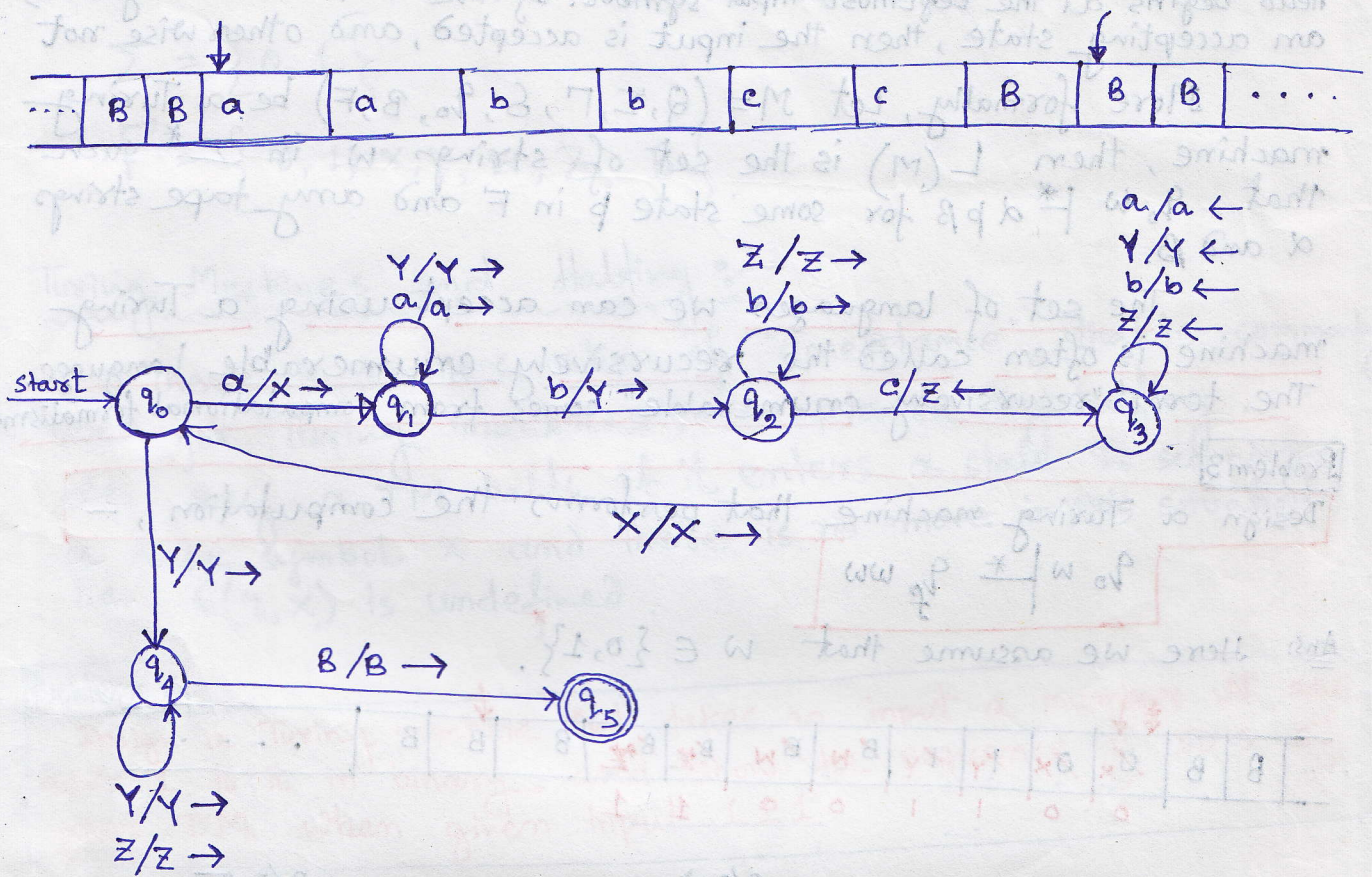
For another example, consider what M does on the input 0010, which is not in the language accepted,

$q_0 0010 \vdash x q_1 010 \vdash x 0 q_1 10 \vdash x q_2 0Y0$
 $\vdash q_2 x 0Y0 \vdash x q_0 0Y0 \vdash x x q_1 Y0$
 $\vdash x x Y q_1 0 \vdash x x Y 0 q_1 B$

However in state q_1 , M has no move on tape symbol B ; thus M dies and does not accept its input.

Problem 2

Design a Turing machine for the language $L = \{a^n b^n c^n \mid n \geq 1\}$



$M = (\{q_0, q_1, q_2, q_3, q_4, q_5\}, \{a, b, c\}, \{a, b, c, x, y, z, B\}, \delta, q_0, B, \{q_5\})$

Transition diagram for a TM that accepts L .

Transition table for the TM to accept $\{a^n b^n c^n \mid n \geq 1\}$.

State	Symbol						
	a	b	c	x	y	z	B
q_0	(q_1, x, R)	—	—	—	(q_4, y, R)	—	—
q_1	(q_1, a, R)	(q_2, y, R)	—	—	(q_1, y, R)	—	—
q_2	—	(q_2, b, R)	(q_3, z, L)	—	—	(q_2, z, R)	—
q_3	(q_3, a, L)	(q_3, b, L)	—	(q_0, x, R)	(q_3, y, L)	(q_3, z, L)	—
q_4	—	—	—	—	(q_4, y, R)	(q_4, z, R)	(q_5, B, R)
q_5	—	—	—	—	—	—	—

The Language of a Turing Machine :-

We have intuitively suggested the way that a Turing machine accepts a language. The input string is placed on the tape and the tape head begins at the leftmost input symbol. If the TM eventually enters an accepting state, then the input is accepted, and otherwise not.

More formally, Let $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$ be a Turing machine, then $L(M)$ is the set of strings w in Σ^* such that $q_0 w \vdash^* \alpha p \beta$ for some state p in F and any tape strings α and β .

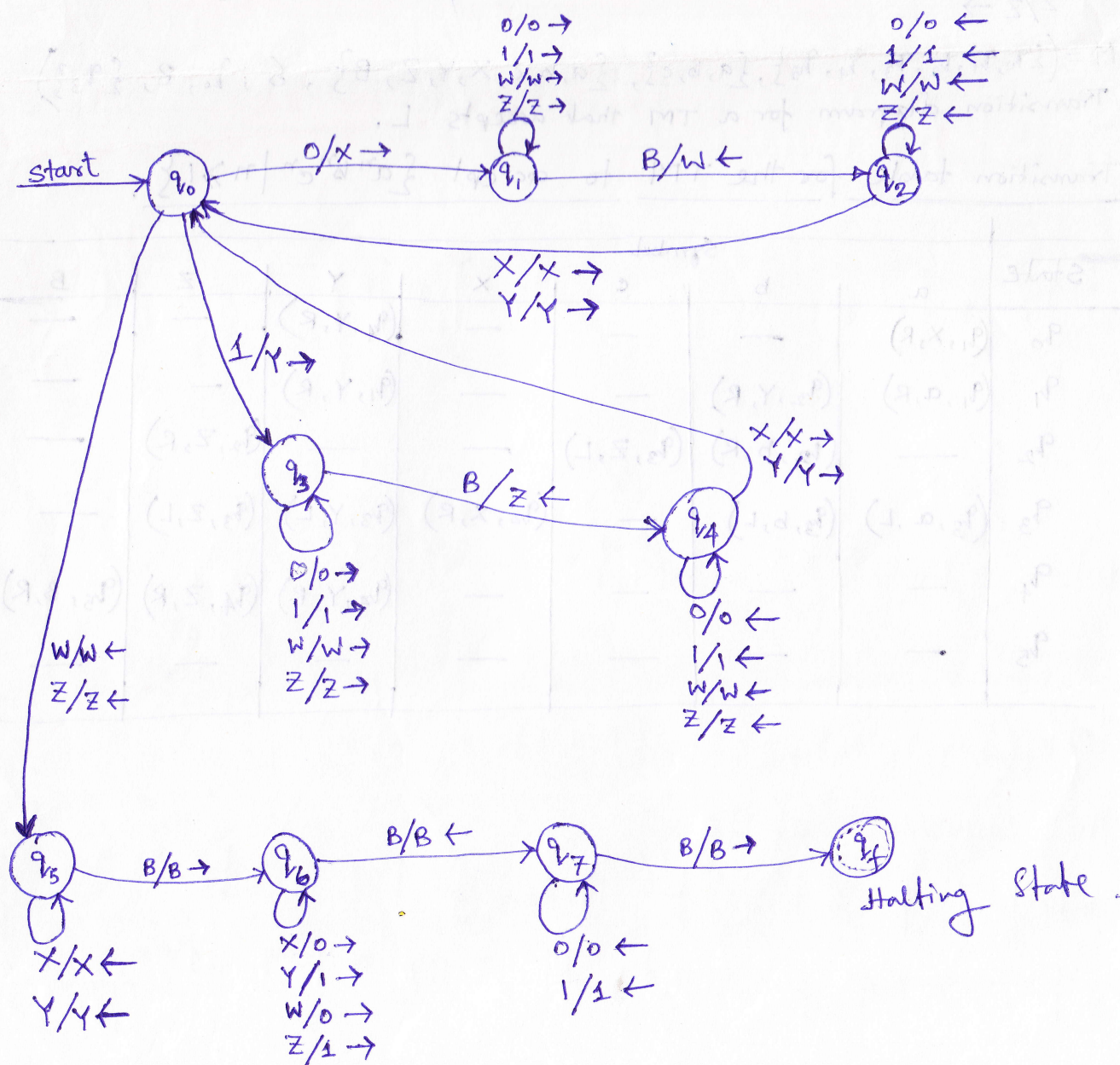
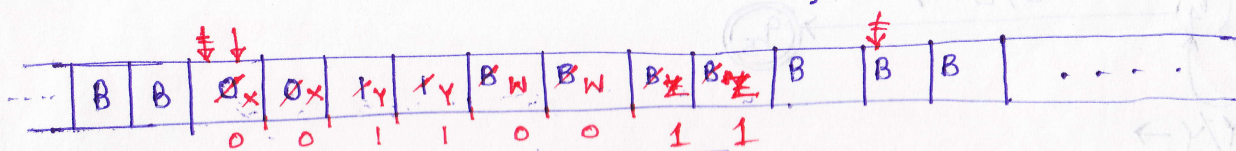
The set of languages we can accept using a Turing machine is often called the recursively enumerable language. The term "recursively enumerable" comes from computational formalisms.

Problem 3:

Design a Turing machine that performs the computation, —

$$q_0 w \vdash^* q_f ww$$

Ans: Here we assume that $w \in \{0, 1\}^*$.



$$M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$$

$$Q = \{q_0, q_1, q_2, q_3, q_4, q_5, q_6, q_7, q_f\}$$

$$\Sigma = \{0, 1\}$$

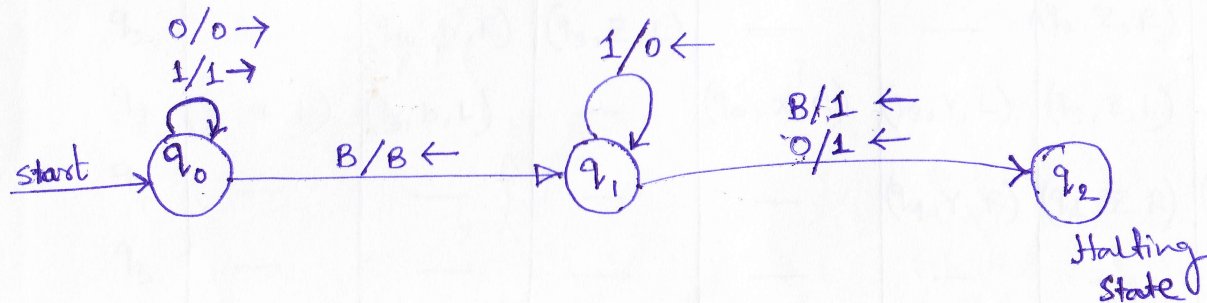
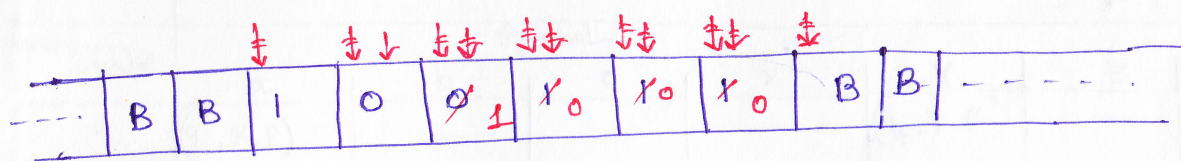
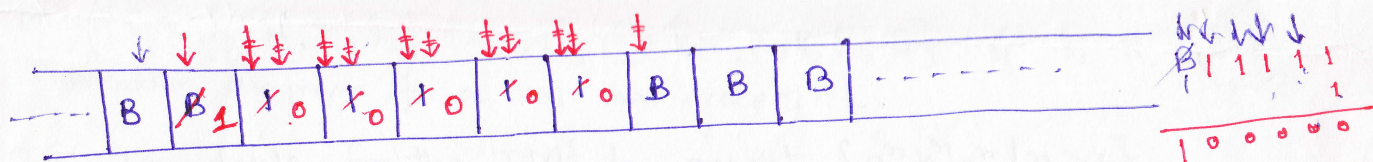
$$\Gamma = \{0, 1, x, y, w, z, B\}$$

Turing Machines and Halting :-

There is another notion of "acceptance" that is commonly used for Turing machines: acceptance by halting. We say a TM halts if it enters a state q_0 scanning a tape symbol x and there is no move in this situation; i.e., $\delta(q, x)$ is undefined.

Problem 4

Design a Turing machine that takes as input a number N and adds 1 to it in binary and show the sequence of ID's of your TM when given input 111



$$M = (\{q_0, q_1, q_2\}, \{0, 1\}, \{0, 1, B\}, \delta, q_0, B)$$

The Sequence of ID's :-

$$q_0 111 \vdash 1 q_0 11 \vdash 11 q_0 1 \vdash 111 q_0 B \vdash 11 q_1 1 \vdash 1 q_1 10 \vdash q_1 100 \vdash q_1 B000 \vdash q_2 B1000$$